

Qatar-1b

R-2672

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160605 · created 2026-08-30T01:19:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457544.78707 +/- 0.00081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.039
Transit depth (Rp/Rs)^2	0.89 +/- 0.74 [%]
Orbital Inclination (inc)	83.38 +/- 2.33
Airmass coefficient 1 (a1)	1.003 +/- 0.013
Airmass coefficient 2 (a2)	-0.0028 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	1.34 %
Best Comparison Star	#2 - [399, 275]
Optimal Aperture	3.79
Optimal Annulus	14.91
Transit Duration (day)	0.054 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.094 $\pm$ 0.039 vs 0.1463 (deviation 1.34 σ)
Duration fit vs published	0.054 d vs 0.0692 d (deviation 0.72 σ)
Mid-time O-C	2.0 min
Depth SNR	2.4
Residual scatter	1.34 %

Light curve

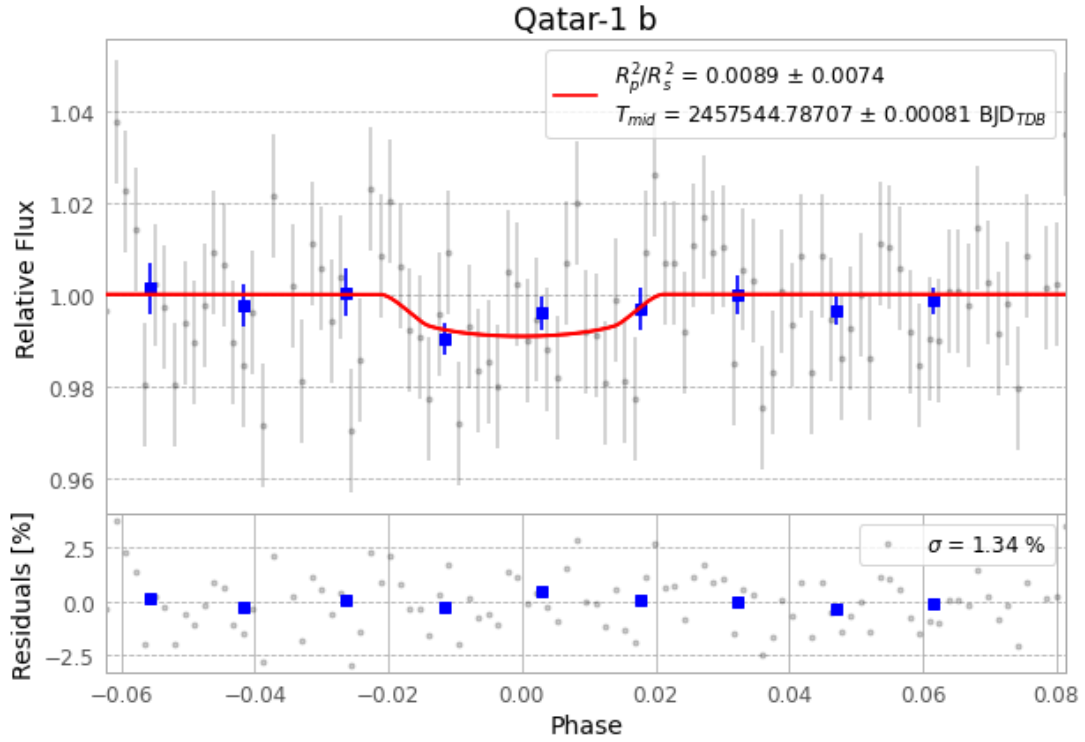


Figure 1 — FinalLightCurve_Qatar-1 b_2016-06-04.png (SHA-256 dcf68d4e41660e3b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31c3788a60cc4a7c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

99 FITS frames (65 MB), Qatar-1160605044512.FITS → Qatar-1160605093913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160605.log (4a080999df42...), FinalLightCurve_Qatar-1 b_2016-06-04.png (dcf68d4e4166...), FinalLightCurve_Qatar-1 b_2016-06-04.csv (a70127eef10b...)

Compute resources

Wall time 143.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:19Z.